

ÖMER SAFA ÇAVUŞ

MANUFACTURING ENGINEER

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PROFILE

Manufacturing engineer completing an Integrated Ph.D. at Sabanci University (defense scheduled for December 2026), specializing in physics-based modeling of laser beam powder bed fusion. Developed an in-house parametric framework that connects part design, scan-path generation, thermal simulation and build-file export, with numerical models validated through X-ray CT, XRD, SEM and mechanical testing. Currently extending this work to beam shaping, machine learning for process monitoring, and other metal additive manufacturing processes such as directed energy deposition (DED), cold spray and electron beam melting (EBM). In industry, carried out additive manufacturing process design, including part-scale thermal analysis, support structure engineering and DfAM-based geometry optimization, and worked across the aluminum casting process, from developing in-house software for automated mold design to heat treatment and material characterization.

EDUCATION

Sabanci University, Ph.D. in Manufacturing Engineering <i>PhD Thesis: Planning and analysis of laser beam powder bed fusion processes supervised by Prof. Dr. Bahattin Koç</i>	2020 – present, Istanbul
Istanbul Technical University, B.Sc. in Civil Engineering <i>Graduation Project: Repairing, protecting and strengthening ageing, damaged or overloaded concrete, and masonry structures with FRCM supervised by Assoc. Prof. Dr. Bekir Yılmaz Pekmezci</i>	2014 – 2020, Istanbul
Hamburg University of Technology, B.Sc. in Civil Engineering <i>Erasmus+ Exchange Program</i>	2018, Hamburg

PROFESSIONAL EXPERIENCE

Machina3D, R&D Engineer <i>Additive Manufacturing: Executed end-to-end process design, including part-scale thermal analysis, support structure engineering, and geometry optimization based on Design for Additive Manufacturing (DfAM) principles.</i> <i>Casting: Developed a specialized in-house software for automated mold generation, and conducted comprehensive research on heat treatment recipes and advanced material characterization methods, with a primary focus on aluminum alloys.</i>	2024 – 2026
Sabanci University Integrated Manufacturing Technologies Research and Application Center (SU-IMC), Researcher <i>Project 1: New Generation 3D Printing Manufacturing Technologies Platform. Conducting research to develop next-generation additive manufacturing technologies and produce high value-added products for the aerospace and defense industries. (Partners: TUBITAK, Sabanci University, ROKETSAN)</i>	2020 – present
<i>Project 2: Development and Experimental Validation of Innovative Topology Optimization Algorithms Supported by Numerical Models. Developed in-house scanning strategy algorithms and computational environments utilizing computer graphics and parametric modeling techniques. Developed in-house multi-physics solvers and implemented process simulations across different scales. Conducted comprehensive material characterization and mechanical testing to validate numerical models. (Partners: TUBITAK, Sabanci University, TEI)</i>	2023 – present
Sabanci University, Teaching Assistant <i>Course 1: Introduction to Computer-Aided Drafting and Solid Modeling with SolidWorks</i> <i>Course 2: Manufacturing Processes I</i> <i>Course 3: Additive Manufacturing</i>	2020 – present 2024 – present

JOURNAL PAPERS

- Khalilvandi Behrouzgar, S., **Cavus, O. S.**, & Koc, B. (2026). A parametric heat source formulation for arbitrary laser beam shaping: application to core–ring profiles and their transient thermal history in PBF-LB. *Submitted to International Journal of Extreme Manufacturing*.
- Cavus, O. S.**, Khalilvandi Behrouzgar, S., & Koc, B. (2026). Prediction of cooling rates based on integrated scanning strategies with sophisticated parametric modeling-assisted simulation for additive manufacturing. *Progress in Additive Manufacturing*.
- Cavus, O. S.**, Khalilvandi Behrouzgar, S., Javidrad, H. R., & Koc, B. (2026). Continuous parametric scanning with thermal modeling to minimize stitch zone defects in laser beam powder bed fusion validated by multi-scale X-ray CT *Additive Manufacturing*.
- Kendibilir, A., Khalilvandi Behrouzgar, S., **Cavus, O. S.**, Lahe Motlagh, P., Koc, B., & Kefal, A. (2026). Enhancing mechanical

- performance of metal additively manufactured structures via peridynamic topology optimization and thermo-mechanical process simulation with experimental validation. *Computers & Structures*.
5. Khalilvandi Behrouzfar, S., Kendibilir, A., **Cavus, O. S.**, Lahe Motlagh, P., Koc, B., & Kefal, A. (2025). Minimizing thermally induced residual stresses in metal additive manufacturing through peridynamics topology optimization. *Engineering with Computers*.
 6. **Cavus, O. S.**, Khalilvandi Behrouzfar, S., & Koc, B. (2024). Functional island scanning strategy for powder bed fusion. *Journal of Additive Manufacturing Technologies*.
 7. Kendibilir, A., Polat, M. F., **Cavus, O. S.**, Behrouzfar, S. K., Motlagh, P. L., Koc, B., & Kefal, A. (2022). Combination of peridynamics and genetic algorithm based topology optimization methods for AM-friendly designs. *Journal of Additive Manufacturing Technologies*, 2(1), 704-704.

CONFERENCE PAPERS

1. **Cavus, O. S.**, Khalilvandi Behrouzfar, S., Javidrad, H. R., & Koc, B. (2026). Investigating the role of scanning strategies on porosity formation and mitigation in stitch regions via thermal modeling and X-Ray CT. *Paper will be presented at the 6th AMCTURKEY Additive Manufacturing Conference, Antalya, Turkey*.
2. Ozyilmaz, P., **Cavus, O. S.**, Khalilvandi Behrouzfar, S., Erkul, I., & Koc, B. (2026). Assessing the limits of in-situ back-scatter electron imaging in narrow electron beam powder bed fusion process windows using X-Ray CT. *Paper will be presented at the 6th AMCTURKEY Additive Manufacturing Conference, Antalya, Turkey*.
3. **Cavus, O. S.**, Khalilvandi Behrouzfar, S., & Koc, B. (2025). Transient and spatial evolution of temperature gradients and solidification rates in laser beam powder bed fusion. *Paper presented at the 5th AMCTURKEY Additive Manufacturing Conference, Antalya, Turkey*.
4. Khalilvandi Behrouzfar, S., **Cavus, O. S.**, & Koc, B. (2025). A comparative study of single and multi-laser beam effects on thermal gradient and solidification rate in LB-PBF. *Paper presented at the 5th AMCTURKEY Additive Manufacturing Conference, Antalya, Turkey*.
5. **Cavus, O. S.**, Khalilvandi Behrouzfar, S., & Koc, B. (2024). Design thermal analysis and optimization of the novel geometry based scan strategies. *Paper presented at the 4th AMCTURKEY Additive Manufacturing Conference, Antalya, Turkey*.

PROJECTS UNDER REVIEW

1. **Horizon Europe Framework Programme, Researcher, Principal Investigator: Prof. Dr. Bahattin Koç**
Project: Next-generation Engineering for Optimized Lifecycle of Critical Materials Supported by AI and Additive Manufacturing to Promote Re-standardization and Circularity
(This proposal will be resubmitted in the upcoming call following the initial review.)
2. **Marie Skłodowska-Curie Actions, Researcher, Principal Investigator: Prof. Dr. Bahattin Koç**
Project: Laser-Based Digital Manufacturing for Energy Efficient Processes and Products
3. **TÜBİTAK-GSRI, Researcher, Principal Investigator: Prof. Dr. Bahattin Koç**
Project: Integrated Additive Manufactured Sensing System for Marine Applications

ENGINEERING SKILLS

- **Additive Manufacturing:** Materialise Magics, Simufact Additive, Netfabb, PBF-LB: EOS M 290, PBF-EB: Freemelt ONE, Cold Spray: Dycomet Titomic PCS-100
- **Finite Element Analysis:** MSC Marc, ANSYS Workbench, User Defined Subroutines
- **Material Characterization and Microstructure:** X-Ray Diffraction Analysis (XRD), X-Ray Computed Tomography (X-Ray CT), Scanning Electron Microscope (SEM), Optical Microscope, SPPARKS Kinetic Monte Carlo Simulator
- **Programming:** MATLAB, Python, C#, Fortran, L^AT_EX
- **Design:** Parametric Modelling, Rhinoceros/Grasshopper, SolidWorks
- **Topology Optimized Based Design:** MSC Apex, Fusion 360, ANSYS Workbench, nTop
- **Other:** AI, Machine Learning & Statistics, Mechanical Behaviour of Materials, Manufacturing Fundamentals, Engineering Optimization

CERTIFICATES & TRAININGS

DIMAP Project Trainings

DIMAP (Direct Digital Manufacturing Platform), co-financed by the European Union and the Republic of Turkey. The project aims to facilitate the implementation of digital production and additive manufacturing methods, particularly in the aviation and automotive sectors.

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| • Directed Energy Deposition & Post Processes , Politecnico di Milano, by Prof. Ali Gökhan Demir | 2025 |
| • Practical Training on Cold Spray , Titomic, by Ferdy Touwen | 2025 |
| • Theory of Cold Spray , Politecnico di Milano, by Prof. Sara Bagherifard | 2025 |
| • Scanning Electron Microscopy , Zeiss, by Volkan Erkut | 2025 |
| • Digital Simulation , Politecnico di Milano, by Prof. Marco Grasso | 2025 |
| • Electron Beam Melting , Freemelt, by Camilo Medina Viramontes | 2025 |
| • X-ray Diffraction (XRD) , Anka Analitik, by Feda Rahal | 2025 |

AWARDS

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| • Performance Scholarships , TUBITAK | 2024 |
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- **Project Scholarships**, TUBITAK 2020
- **High Honour List**, Istanbul Technical University 2018
- **Grant for Erasmus+ Exchange Students**, European Union 2018
- **A2 German Language Certificate**, Goethe Institut 2017
- **Certificate of Appreciation**, UNICEF 2017

LANGUAGES

Turkish (Native), English (Fluent), German (Elementary)

INTERESTS

Skiing, Scuba Diving, Travelling

REFERENCES

Prof. Dr. Bahattin Koç Sabanci University	Assoc. Prof. Dr. Adnan Kefal Sabanci University	Selim Dinçer-Co Founder Machina3D	Dr. Utku Seven Sabanci University
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